Interpreting and Coding Open Responses

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Coding and Classification: Outline

What are open responses?

Characteristics of everyday classification

- models of categorization; hierarchical structure; graded structure

Everyday classification may contribute error to coding process

- squeezing responses into categories where they may not fit (Kunda & Oleson)
- mistaking typicality for probability (Tversky & Kahneman)

Studies of coding

- Adding psychologically meaningful categories (Malhotra & Krosnick)
- Coder rules that improve reliability but not validity (Hak & Berndt)
- Length of description may affect quality of coding (Conrad, Couper & Sakshaug)
- Coder variance: field versus office coding (Martin et al.); impact of coder clustering (Sturgis)
- Rigid coding can lead to over-exclusion (Gibson & Caldiera)

Open Responses

- Some survey questions ask *R*s to answer in their own words, i.e., no response options are provided
  - "Can you share more about your experiences using our university library?"
  - "What motivated you to choose your current profession or field of study?"
  - "You said you have not registered to vote. Please tell us why you have not registered to vote?"
- So that open responses can be quantified, need to be assigned to a formal category, i.e., *coded* or *classified*
  - "I have not registered to vote because my vote won't make a difference."
  - assign to "Disbelief in System," "Felony Conviction," "Lack of Compelling Candidates," "Discontent with Politics," "Privacy Concerns,"…

Open Responses: The Case of Occupation

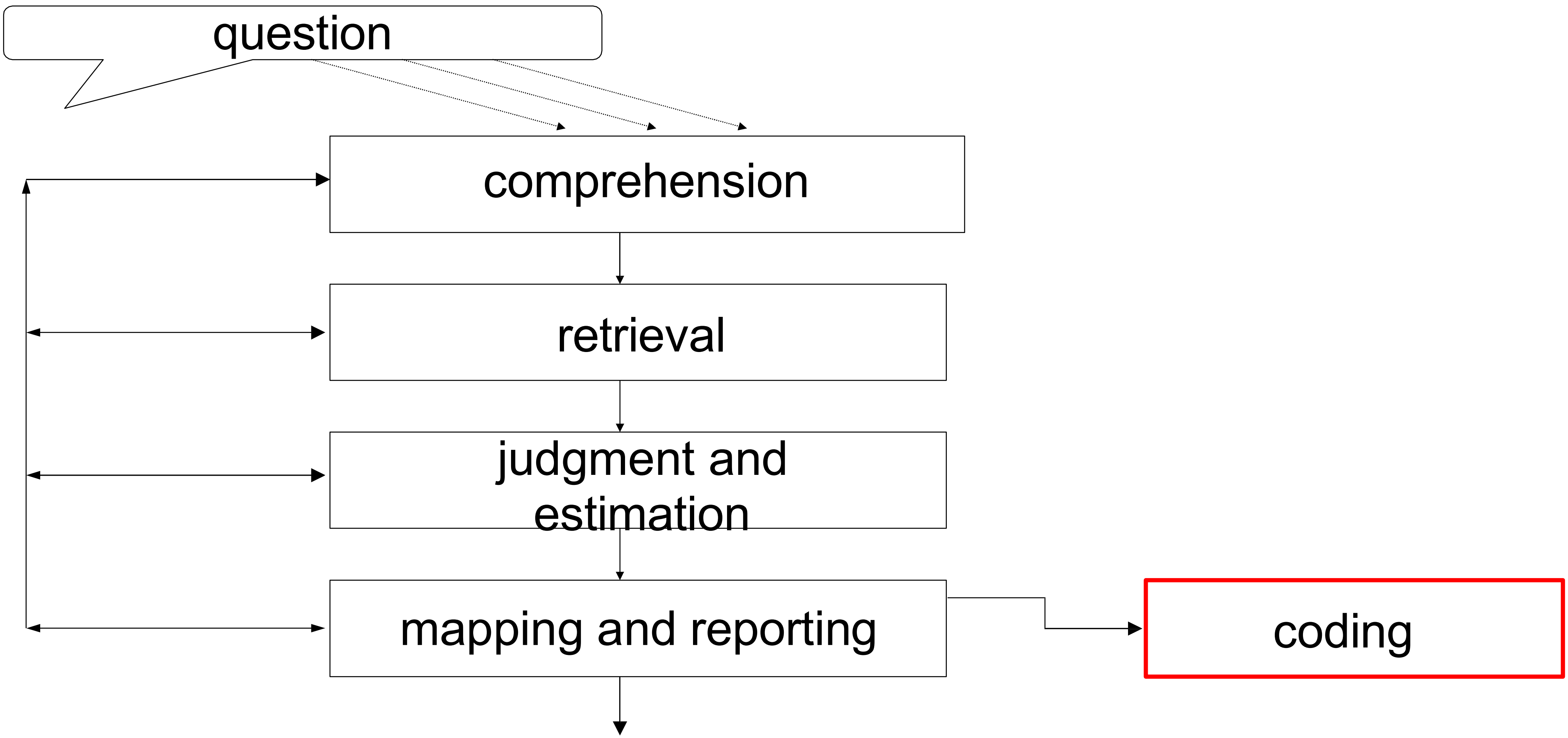
- Current Population Survey asks two questions about *R*'s job

1. What kind of work do you do, that is, what is your occupation?
2. What are your usual activities or duties at this job?

- *R*s answer in their own words, transcribed by *Iwer*, coded by professional coders (and computers) into one of > 800 occupational categories
- Both responses considered together in assigning a numerical code to *R*'s job
  - US Census occupational classification system
    - 23 major groups (1 digit); 93 minor groups (2 digits); 509 categories (3 digits)
- Responses form basis of important official statistics
- Would be unreasonable to ask *R* to assign their job to one of 500+ categories

Open vs. Closed Questions

- Open questions may elicit answers that would not have been produced if fixed options presented
  - Schuman & Presser (1981) asked about most pressing problem facing country in open or closed form
    - 22% of open responses concerned "energy shortage" (which had just become a major issue)
    - Closed form did not include "energy shortage"
- But specific options can remind people of possible answers they would not think of with open form
  - Shuman & Presser (1981) asked what attributes of a job people would prefer
    - Open version led predominantly to answers about "pay" though could not distinguish high from steady pay
    - Closed form presented options for both: "work that pays well" vs. "work that is steady" allowing researchers to distinguish between the two senses of "pay"



- Coding error can be thought of as measurement error: a human judgment leads to a discrepancy between what is observed (the code) and the true value (assuming this can be determined); sometimes considered “processing error.”

Three Models of Everyday Categorization

(inspired by Barsalou, 2014)

Exemplar model:

- People compare a to-be-classified entity to remembered exemplars (members) of individual categories
- the new entity becomes a member of the category which includes an exemplar most *similar* to the new entity

Prototype model:

- people compare a to-be-classified entity to an abstraction – a prototype – of categories
- the new entity becomes a member of the category whose prototype is most *similar* to a description of the new entity

Classical model:

- people apply <u>rules</u> for category membership to a new entity
- rules specify necessary and sufficient properties for membership
- if rule matches new entity, it becomes a member of that category

Coders and Models of Classification

- Coders classifying open responses likely follow rules, much as in Classical model
- but may also apply exemplar and prototype models when rules are not available or do not sufficiently discriminate among categories

Everyday Categorization: Hierarchical, graded structure, flexible

- Most theories of categorization assume that more abstract categories entail more concrete categories, i.e., that natural categories are hierarchical in structure
  - *animals* includes mammals, birds, fish, insects ...
    - *mammals* includes *dogs*, *cows*, *rodents* …
      - *dogs* includes *spaniels, retrievers, poodles ...*
        - » *retrievers* includes *Golden, Chesapeake, Labrador …*
- People prefer categories at intermediate levels of abstraction -- <u>basic level</u> (Rosch, e.g.,1973)
  - "What are you sitting on?" People prefer to answer "chair" than "furniture" or "kitchen chair"
  - "performance," e.g., speed of classification, generally worse if category more abstract (superordinate) or more concrete (subordinate)

Everyday Categorization: Hierarchical, graded structure, flexible (2)

- Under exemplar and prototype models, entities can belong to a category to varying degrees
  - graded structure or *typicality* affects classification
    - A robin is a more typical than is an ostrich
    - People are faster to decide a robin is a bird than that an ostrich is a bird
  - category boundaries are flexible; adjusted based on new members
- Classical model (which is implicitly used by coders) does not permit graded membership
  - an instance is either in or out of the category

Coding: Reliability, Validity, and True Value

- When an open response is coded, there may not be a single correct judgment, i.e., a true value
  - If *R*'s behavioral report is in question (e.g., voted in last election?), voting records or other documentary evidence could, in principle, resolve the issue
  - But no such resolution is available for assigning a verbal description to a category, e.g., is it correct that ”I don't vote because politicians are corrupt” is an instance of “Discontent with Politics” or should it be assigned to ”Disbelief in the System?”
- Makes it hard to discuss true value, and thus validity, of codes
- *Intercoder reliability* is the typical measure of coding quality
  - easy to compute
  - but can be high when validity is low, *i.e.,* coders can agree with each other and be “wrong”
- Expert judgment, e.g., a supervisor may reconcile conflicting codes, but really a matter of judgment, i.e., no gold standard

Coding: Classifying atypical instances

- What happens when an instance seems to belong to a category but differs from other members on a single attribute?
- Exemplar and prototype models would allow graded membership
  - atypical case at the edge of category boundary
- Classical model makes no typicality distinction
  - requires people to:
    - change thinking about definition of the category, or
    - create subcategory for atypical cases

Are these all *pens*?

Creating Subcategories to Accommodate Atypical Instances

Kunda and Oleson ’95 (Exp. 1)

- Introverted (atypical) lawyer, Steve, presented in fictional interview transcript
- *P*s rate introversion of lawyers in general (the category) after exposure to Steve’s interview
- Some *P*s given extra information about the law firm e.g., large or small, to justify creating a subcategory (e.g., introverted lawyers)
  - Prediction: Steve’s atypicality should <u>not</u> affect introversion ratings of *lawyers* in general if it is possible to create subcategory for Steve, i.e., introverted lawyer
- Other participants given no information to justify subtyping
  - Prediction: Steve’s atypicality <u>should</u> affect ratings of *lawyers* (more introverted) because atypical target incorporated into category
- Control participants rated lawyers without reading “interview” transcript so did not know Steve was introverted

Creating Subcategories (2)

- Impact on overall ratings of lawyer introversion depends on ability to create subcategories

Control	No info.	Small firm	Large firm
3.73	4.46	3.71	3.68

Introversion Ratings for Lawyers in General (1= not at all, 11=extremely)

- So, in daily life, people adjust definition of categories by creating subcategories for atypical instances
  - e.g., black squirrels are squirrels, just not grey
- What happens to <u>coders</u> who cannot adjust the definition of the category or create subtype (e.g., add 5th digit) for atypical case?

Mistaking typicality for probability

Tversky and Kahneman's **representativeness heuristic** leads to insensitivity to base rate info:

- someone who seems like an engineer, based on a verbal description, is more likely to be judged an engineer than a lawyer *even if there are more lawyers in the "sample"*

T&K (1983) demonstrated a more extreme confusion of typicality and probability (frequency) -- the **conjunction fallacy**:

- someone (Linda) who is judged to be *typical* of a conjunctive category like "feminist bank teller," based on a verbal description, is judged more *likely* to be a member of that category than of the constituent category "bank teller" *even though there are more bank tellers than feminist bank tellers*

- If people are insensitive to base rates and give priority to typicality in everyday thinking, then coders may do this too

Extending the coding system

- If descriptions do not fit existing categories
  - New categories can sometimes be added
  - The entire coding system may be updated, e.g., Standard Occupational Classification system (SOC) is generally updated every ten years before decennial census
- Malhotra & Krosnick (2007) present a procedure to update coding systems so that new categories are psychologically meaningful
  - should promote easier and more reliable coding

Malhotra & Krosnick, 2007

- Goal: extend Standard Occupational Classification (SOC) system to accommodate new and growing job categories
  - *Biotechnology* is test bed
- Domain experts asked to either
  1. Classify titles of new jobs into existing SOC and time them
     - Provide judgments of *fit* and *confidence*
  2. Sort similar titles into similar (consensual) piles and label piles

Classification time (sec.)
for judgments of “Fit” and “Confidence”

	Fit	Confidence
5	28.93	32.68
4	32.86	34.46
3	45.43	44.06
2	50.41	49.97
1	63.95	88.14

5 is best, 1 is worst

Accommodating instances that do not fit current categories

- 40 jobs had composite score* more than one σ below mean
- These were not in current SOC
- Of these, 21 fell into "consensual piles" from sorting task:
  - Clinical, QA/QC, Manufacturing, Regulatory, Project Manager
  - i.e., coherent categories not in SOC
- These are the proposed new categories

*consists of classification speed, agreement, fit, and confidence. Smaller scores => less correspondence to a job in SOC

Socially derived coding rules

- Hak & Bernts (1996) observe that coders jointly develop rules to handle cases not covered by the instructions
  - "I think that … if the phrase 'being able to' occurs in the answer we can almost always choose VALUE if it is possible uh .. to read the answer in that way
- Development of informal rules may be triggered when terms in *R*s' descriptions fit poorly into existing categories or fit well into > 1 category

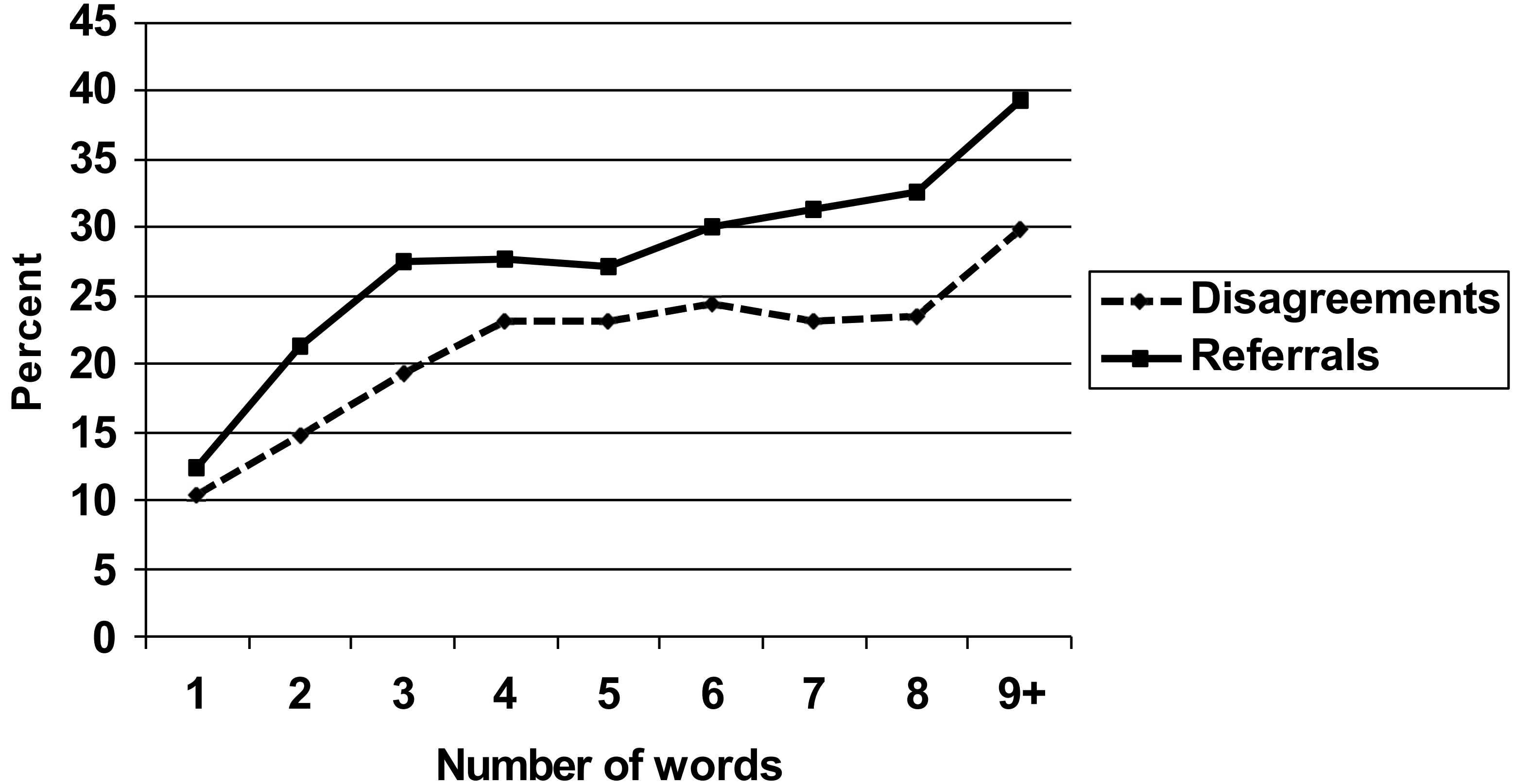
Socially derived coding rules (2)

- This kind of rule may improve reliability but is unrelated to validity (accuracy) of classification
  - Hak & Bernts observed that when coders agreed, they never discussed their reasons for agreement
  - coders follow rule of economy: if it is possible to settle on a single code, then do so
  - Possible to agree for the wrong reasons so both coders are "wrong," and another code is actually "right"
- Informal coding rules lead to lower agreement because they
  - are undocumented so are not consistently followed, and
  - are ad hoc and atheoretical, so may conflict with one another

Length of Open Response and Coding Agreement

- Conrad, Couper & Sakshaug (2016) conducted 3-part study of occupation coding in Current Population Survey
  - (1) What characteristics of occupation descriptions hurt coding reliability?
    - Observational study: double-coded descriptions (n=32,362) created for QA purposes
  - (2) How do characteristics jointly affect reliability?
    - Experiment: double-coded descriptions differ in length and difficulty (n=800)
  - (3) What do coders think about while coding?
    - Coders think aloud while classifying experimental descriptions (n=100); their think-alouds analyzed for use of informal rules
- Found in (1) that disagreement is higher for longer descriptions but in (2) effect of length depends on difficulty of terms in *R*s' descriptions
- Use of informal rules evident in part (3)

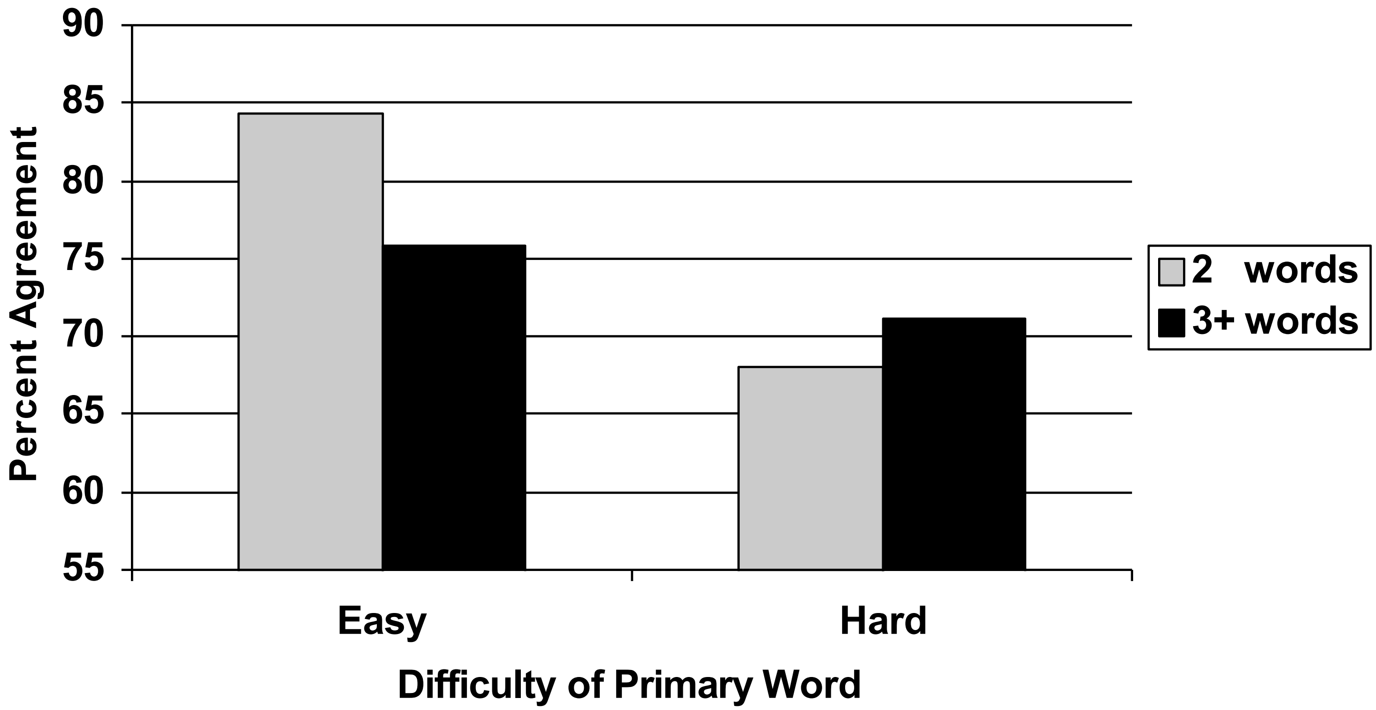
Length of description and disagreement/referral rates



Joint effects of difficulty and length

- In subsequent experiment, descriptions selected to systematically differ in:
  - length (1, 2 and 3+ words)
  - difficulty (easy, hard)
    - easy=highest ratio agreement to disagreement in QA data:
      secretary, cashier, driver, cook, teacher, nurse, waitress, carpenter
    - hard=highest ratio disagreement to agreement in QA data:
      owner, operator, laborer, director, technician, clerk, supervisor, administrator

Increased length lowers agreement when words are *easy*



Special purpose rules: More than one occupation

- In observational study, all (4) coders reported using special purpose rules
  - concerned superficial aspects of descriptions, e.g., “two jobs described”
  - not definitions of coding categories (atheoretical)
- None of the coders could produce the rules in writing

Rule: “When two different occupations are described, match to ‘duties’ and go with the first duty listed.”

```
EMP     : DOMINOS PIZZA
IND     : PIZZA
OCC     : COOK, DRIVER
DUTIES  : DELIVERY, COOKING
```

Special purpose rules:
More than one occupation (2)

"Rule: When two different occupations are described, match to 'duties' and go with the first duty listed."

```
EMP    : NEWBERRYS
IND    : VARIETY STORE
OCC    : CASHIER  STOCKING
DUTIES : STOCKING AND CASHIER
```

Coding during data collection

- Martin, Bushnell, Campanelli & Thomas (1995):
  - compared occupation coding by conventional (office) coders and interviewers (field coders)
  - No advantage in “accuracy” for field (interviewer) vs. office (clerical) coding: 77% vs. 80% agreement with expert
  - But between-coder *reliability* (agreement) is consistently *lower* for field than office coders
  - Lower agreement may indicate better quality coding by field than office coders
  - because of (1) larger work force, and (2) lower individual workload:
    - (1) and (2) reduce correlated error (clustering by coder) for field coders compared to office coders by diluting effect of any one coder
    - also suggests field coders not able to develop informal rules as are office coders because of latter’s proximity to each other and more contact

Idiosyncratic Coder Behavior

- If some coders consistently assign descriptions to different categories than their colleagues, someone is wrong
- Sturgis (2004) examined this by applying statistical procedures quantifying logic of *interviewer effects* to coders in UK Time Use Study:
  - The more variance due to coders, the less precise the estimates, i.e., standard error of estimates (*ceft*) is inflated
  - Coding staff (n=5) classified descriptions in 40 time-diaries
  - Extrapolated from these cases to full sample of ~21,000
  - When scaled up to full sample, very large workload (~3000 diaries per coder) magnifies small idiosyncrasies, leading to large inflation of standard error

Bias Versus Variance

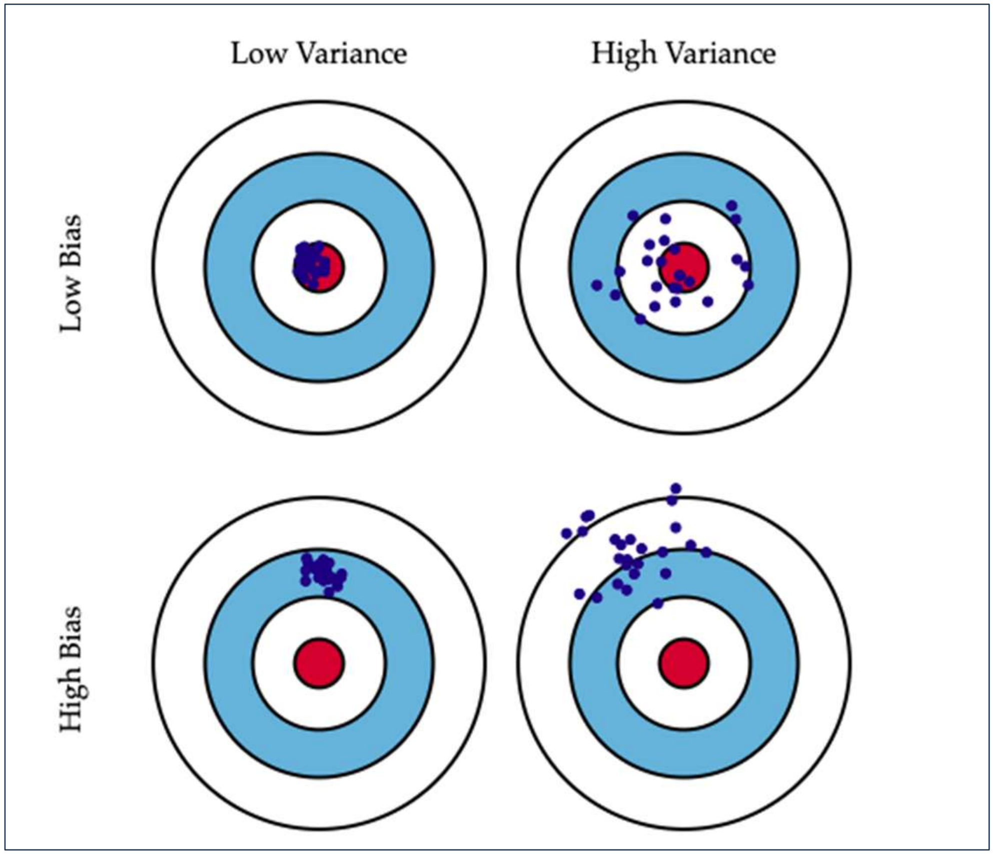
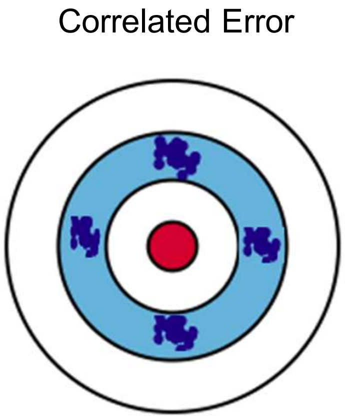


Image source: https://scott.fortmann-roe.com/docs/BiasVariance.html



Variance Inflation Due to Clustering within Coder Workloads

ceff = (1+p(m-1)*(1-K)) [square root (ceft) is inflation factor for standard errors]

p = intraclass correlation of coders
m = average coder workload
K = reliability of code

ceff = (1+.02(1000 - 1)*(1 - .75)) = 6

ceff = (1+.01(1000 - 1)*(1 - .75)) = 3.5

ceff = (1+.02(1000 - 1)*(1 - .90)) = 3

ceff = (1+.02(500 - 1)*(1 - .75)) = 3.5

Impact of coder error on precision of estimates

Activity	Time of Day	Point Estimate	S.E. without coder error*	S.E. with coder error
Personal Care	08:20-08:30	52.4%	0.34	0.42
Employment	14:40-14:50	18.7%	0.27	0.40
Study	11:20–11:30	7.3%	0.18	0.30
Household/family care	16:50-17:00	19.5%	0.27	0.24
Voluntary work	15:30-15:40	2.1%	0.10	0.44
Social Life	22:30-22:40	3.5%	0.13	0.10
Sports/outdoor activities	13:20-13:30	12.0%	0.22	0.75
Hobbies and games	16:20-16:30	5.2%	0.15	0.52
Media Consumption	21:20-21:30	43.4%	0.34	0.61
Travel and Unspecified	16:00-16:10	16.6%	0.26	0.71

*assumes simple random sample (no clustering) and 0 correlated coder error

- inflated SEs could lead to overconfidence in estimates of time use
- doubling # coders would have reduced variance inflation by 25-35%

Coding Open Responses to Knowledge Questions

"Now we have a set of questions concerning various public figures. We want to see how much information about them gets out to the public from television, newspapers and the like…. What about … William Rehnquist – What job or political office does he NOW hold?"*

American National Election Survey (2004)

- Gibson & Caldeira (2009) coded these open responses
- Only 12% correct!
- G&C found many acceptable answers coded as "incorrect" because did not include: "Chief Justice" and "Supreme Court"

*Correct answer: Chief Justice of US Supreme Court

Coding Open Responses to Knowledge Questions (2)

- 30% identified Rehnquist as a "Supreme Court justice," but coded as "incorrect"
- 25% either said that Rehnquist was a "judge" or that he was on the Supreme Court and were coded as having answered incorrectly

- G&C proposed closed form *recognition* question as an alternative
- Should leads to higher estimates of knowledge
- *R*s asked to state whether Rehnquist, Lewis F. Powell, or Byron R. White was Chief Justice
  - 71% correctly selected Rehnquist
- Conclusion: Rigid inclusion criteria criteria led to substantial misclassification

Summary and Conclusions

- Coding and classifying open reports is prone to error because
  - *Everyday classification* is similarity-based, i.e., graded and continuous but *coding* is in-or-out
  - Coders may rely on (fall prey to) similarity-based heuristics (representativeness heuristic, conjunction fallacy)
  - Coders cannot modify category definitions to accommodate atypical instances, possibly increasing disagreement
  - Longer open descriptions are more ambiguous (and lower coding agreement); informal rules for longer descriptions may also lower agreement
  - Coder idiosyncrasies are magnified by large workloads
  - Coding rules can be rigid leading to exclusion of accurate descriptions

Summary and Conclusions (2)

- But … open responses are essential, especially when there are too many categories for *Rs* to use
- Improving the quality of open response data is a largely unexplored frontier for methodologists and applied psychologists
  - New classification systems can be designed so categories are intuitive
- NLP (e.g., Topic Modeling) and Large Language Models used increasingly for automated coding
- To the extent these rely on human coding for training, they are subject to same kinds of errors as are human coders